Weighted N6 exponent: formal scaling boundary

aburan28/crypto research packet

25 October 2026

The published genus-2 complete-law article defines a weighted coordinate action. Its companion author-code transcription uses one power of W in the $N6$ intermediate; the frozen Rust runtime uses W^2. The existing $p = 739$ regression gives an input where the W^2 result agrees with independent Cantor arithmetic and the W^1 result does not. This packet proves the coordinate-scaling distinction, not complete Jacobian addition.

Under $(S, T, W) \mapsto (\mu S, \mu T, \mu W)$ with Q, R, Z fixed, the represented affine divisor is unchanged. Put $N1 = QS - T$, $N2 = RS$, $N5 = S^2 - HW^2$ and $N6 = DW^2$. Then both $AA = N1 \cdot N5 - N2 \cdot N6$ and $CC = S \cdot N5 - T \cdot N6$ scale by μ^3. If $N6 = DW$ instead, the exact defects are $\mu^2(\mu - 1)RSDW$ for AA and $\mu^2(\mu - 1)TDW$ for CC. Four universal polynomial identities and a concrete integer counterexample were checked by Lean 4.34.1/Mathlib. The axiom audit lists only `propext`, `Classical.choice`, and `Quot.sound`. The four native identity certificates are `IDC1h92ee5de5110623fd`, `IDC1h7679e312ee733fcb`, `IDC1h87fbf5497ad4d6e0`, and `IDC1he881917e142d1cb5`; each passed 32 sealed and 32 checker-chosen points.

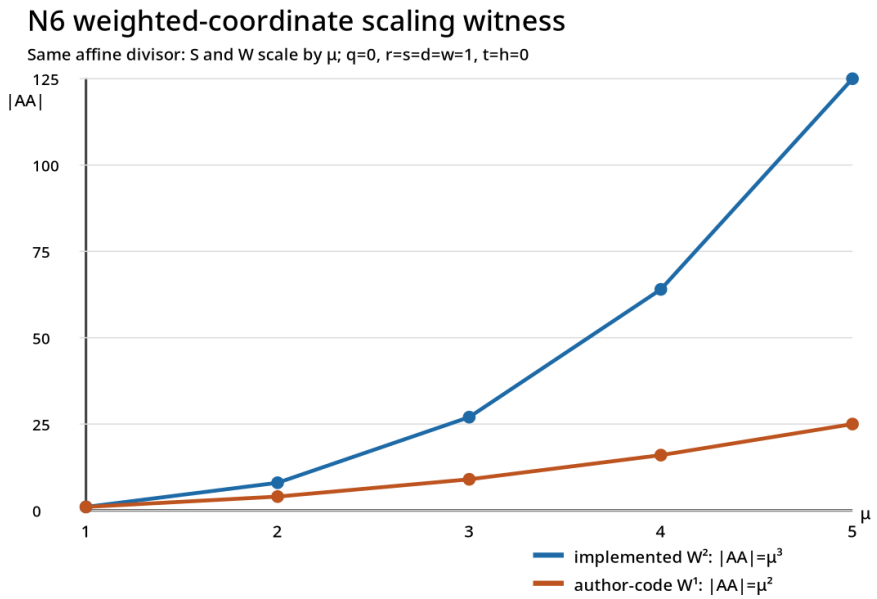


Figure 1: Exact integer illustration from `scaling.csv`: $Q = T = H = 0$, $R = S = D = W = 1$. At $\mu = 3$, the same affine input yields $AA = -27$ for W^2 and $AA = -9$ for W^1. This is a scaling witness, not a cost graph.

The earlier full 1,260-target operation-price ratios are unchanged: wrapper/rho 3.793783 and retained-state/rho 4.179297 at the representative cell. The Rust-to-Lean semantic correspondence, concrete smooth-curve/Mumford/ Pic^0 bridge, and independent mathematical review remain open.

Sources: published article and author code.